2020 AMC 10B Problem 22 - Solution

Review the full statement and step-by-step solution for 2020 AMC 10B Problem 22. Great practice for AMC 10, AMC 12, AIME, and other math contests

2020 AMC 10B Problem 22

Problem:
What is the remainder when $2^{202} + 202$ is divided by $2^{101} + 2^{51} + 1$?

Answer Choices:

A. 100
B. 101
C. 200
D. 201
E. 202

Solution:

$$\begin{aligned} 2^{202} + 202 &= \left(2^{101}\right)^2 + 2 \cdot 2^{101} + 1 - 2 \cdot 2^{101} + 201 \\ &= \left(2^{101} + 1\right)^2 - 2^{102} + 201 \\ &= \left(2^{101} - 2^{51} + 1\right)\left(2^{101} + 2^{51} + 1\right) + 201 \end{aligned}$$

Thus, we see that the remainder is surely $(\mathrm{D}) \boxed{201}$.

$$\textbf{OR}$$

Let $x = 2^{50}$. We are now looking for the remainder of

$$\frac{4x^4+202}{2x^2+2x+1}$$

We could proceed with polynomial division, but the numerator looks awfully similar to the Sophie Germain Identity, which states that

$$a^4+4b^4=\left(a^2+2b^2+2ab\right)\left(a^2+2b^2-2ab\right)$$

Let's use the identity, with $a=1$ and $b=x$, so we have

$$1+4x^4=\left(1+2x^2+2x\right)\left(1+2x^2-2x\right)$$

Rearranging, we can see that this is exactly what we need:

$$\frac{4x^4+1}{2x^2+2x+1}=2x^2-2x+1$$

So

$$\frac{4x^4+202}{2x^2+2x+1}=\frac{4x^4+1}{2x^2+2x+1}+\frac{201}{2x^2+2x+1}$$

Since the first half divides cleanly as shown earlier, the remainder must be $\mathrm{(D)}\boxed{201}$.